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# Rapid Spanning Tree Protocol

## Spanning Tree versions

| | Industry Standards (IEEE) | Cisco Versions |
|---|---|---|
| I) | **Spanning Tree Protocol (802.1D)**<br>• The original STP<br>• All VLANs share one STP instance<br>• Therefore, cannot load balance. | **Per VLAN Spanning Tree Plus (PVST+)**<br>• Cisco's upgrade to 802.1D<br>• Each VLAN has its own STP instance<br>• Can load balance by blocking different ports in each VLAN. |
| ~~II~~ | ~~Re~~ | |
| II | **Rapid Spanning Tree Protocol (802.1w)**<br>• Much faster at converging/adapting to network changes than 802.1D<br>• ALL VLANS share one STP instance<br>• Therefore cannot load balance. | **Rapid Per VLAN Spanning Tree Plus (Rapid PVST+)**<br>• Cisco's upgrade to 802.1w<br>• Each VLAN has its own STP instance.<br>• Can load balance by blocking different ports in each VLAN. |
| III. | **Multiple Spanning Tree (802.1S)**<br>• Uses modified RSTP on mechanics<br>• Can group multiple VLANs into different instances (ie VLANs 1-5 in instance 1, VLANs 6-10 in instance 2) to perform load balancing. | • Cisco hasn't developed their own version of MSTP, Cisco devices simply run the in industry standard (802. |

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For large networks, it is good to use MSTP & for small & medium Rapid-PVSTP+ works well.

## Cisco's Summary:

"RSTP isn't a timer-based spanning tree algorithm like 802.1D. Therefore, RSTP offers an improvement over the 30 sec or more than that 802.1D takes to move a link to forwarding. The heart of protocol is new bridge-handshake mechanism, which allows ports to move directly to forwarding."

## Similarities between STP & RSTP

- RSTP serves the same purpose as STP, blocking specific ports to prevent layer 2 loops
- RSTP elects a root bridge with the same rules as STP
- RSTP elects root ports with same rules as STP.
- RSTP elects designated ports with same rules as STP.

## # Some Differences between RSTP & RSTP.

(1) Cost

| Speed | STP Cost | RSTP Cost. |
|---|---|---|
| 10 Mbps | 100 | 2,00,000 (2 million) |
| 100 Mbps | 19 | 20,0000 (200k) |
| 1 Gbps | 4 | 20,000 |
| 10 Gbps | 2 | 2000 |
| 100 Gbps | 1 (Probably) | 200 |
| 1 Tbps | 1 (Probably) | 20 |

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# Rapid Spanning Tree Port States

In STP there are 5 port states, a) blocking b) listening c) learning d) forwarding e) disabled.

In RSTP there are only 3. Blocking, listening & disabled are combined to one (actually blocking & disabled and listening isn't used). The combined state is called discarding.

| STP Port State | Send/Receive BPDUs | Frame forwarding (regular traffic) | MAC address learning | Stable/ Transitional |
|---|---|---|---|---|
| Discarding | NO/YES | NO | NO | Stable |
| Learning | YES/YES | NO | YES | Transitional |
| Forwarding | YES/YES | YES | YES | Stable |

- If a port is administratively disabled (shutdown command) = discarding state.
- If a port is enable but blocking traffic to prevent layer 2 loops = discarding state.

# Rapid Spanning Tree Port Roles

- In RSTP all switches produce BPDUs, not just root bridge.

- The root port remains unchanged in RSTP
  - ↳ The port that is closest to the root bridge becomes the root port for the switch.
  - ↳ The root bridge is the only switch that doesn't have a root port.

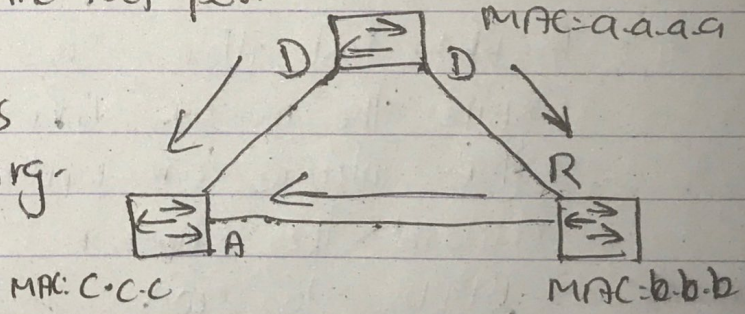
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- The designated port remains unchanged in RSTP.
  ↳ The port on a segment (collision domain) that sends the best BPDU is that segment's designated port (only ~~per~~ one per segment).

- The non-designated port role is split into two separate roles in RSTP.
  the alternate port role.
  the backup port role.

# RSTP: Alternate Port role.

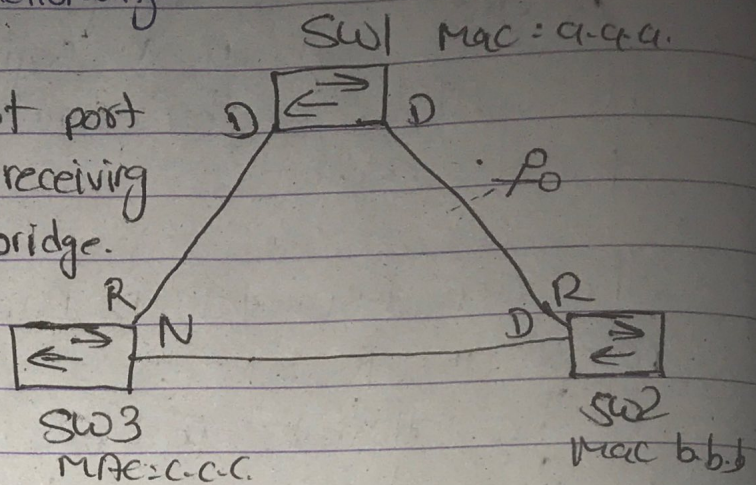
- The RSTP alternate port role is a discarding port that receives a superior BPDU from another switch.
- This is the same as what we've learned about blocking ports in the classic STP.
- Functions as a backup to the root port.
- If the root port fails, the switch can immediately move its best alternate port to forwarding.




This immediate move to forwarding state fxns like a classic STP optional feature called UplinkFast. Because it's built into RSTP, we don't need to activate Uplinkfast when using RSTP/Rapid PVST+.

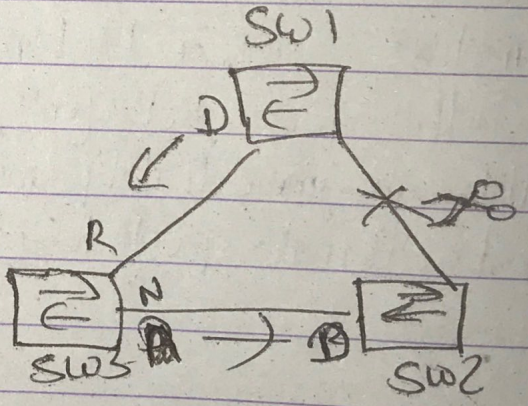
(So अगर current Alternate Port तो backup है Root port का तो If root port fails alternate port'll be root port)

# RSTP - Backbonefast functionality




Let's say SW2's root port is cut off, so it stops receiving BPDUs from the root bridge. Now it assume, it is root bridge & start sending BPDUs to SW3. As SW3 receives BPDUs from SW1 & SW2.

Without backbonefast functionality, SW3 would ignore BPDUs from SW2 until its non-designated port in classic STP finally changes to forwarding state, and starts forwarding superior to BPDUs to SW2. SW2 now accepts SW1 as root bridge.




- Backbone fast allows SW3 to expire the max-age timer on that interface and rapidly forward the superior BPDUs to SW2.

- This fxn is built into RSTP, so it doesn't need to be configured.

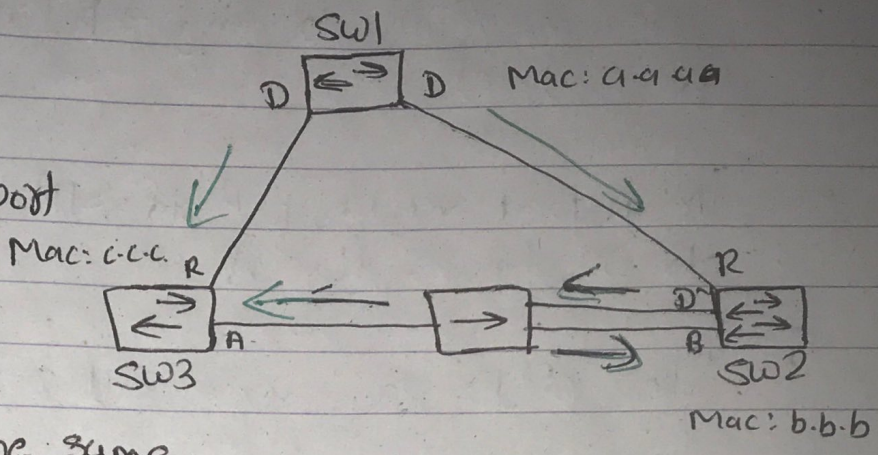
# Uplinkfast / Backbonefast summary

- Both features are built into RSTP, so we don't have to configure them. They operate by default.

# RSTP: Backup port Role.

- ~~This only happens~~

- The RSTP backup port role is a discarding port that receives a superior BPDU from another interface on the same switch.




- This only happens when two interfaces are connected to the same collision domain (via a hub)
- Hubs aren't used in modern networks, so we'll probably not encounter an RSTP backup port.
- Function as a backup for a designated port.

The interface with the lowest port ID will be selected as the designated port, and the other will be backup port).

# Rapid Spanning tree protocol

SW3(config)# spanning-tree mode ?
SW3(config)# # spanning-tree mode rapid-pvst.

Rapid STP is compatible with a classic STP. The interface(s) on the Rapid STP-enabled switch connected to the

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classic STP-enabled switch will operate in classic STP mode (timers, blocking → listening → learning → forwarding process, etc)

- The protocol version of Rapid STP Bpdu is ~~0x02~~ 2.
  BPDU type is (0x02).
- The BPDU flags uses all 8 bits

NOB:

In classic STP, only the root bridge originated BPDUs, and other switches just forwarded the BPDUs they receive.

In rapid STP, all switches original and send their own BPDUs from their designated ports

# Rapid STP.

- All switches running Rapid STP send their own BPDUs every hello time (2 seconds).
- Switches 'age' the BPDU info much more quickly. In classic STP, a switch waits 10 hello intervals (20 seconds). In rapid STP, a switch consider a neighbor lost if it misses 3 BPDU (6 seconds). It will then "flush" all MAC addresses learned on ~~that~~ interface

# RSTP Link Types

Rstp distinguishes between three different 'link types'.

~~Edge:~~

a) Edge: a port that is connected to end host. Moves directly to forwarding, without negotiation.

b) Point-to-Point: a ~~connecti~~ direct connection between two switches

c) Shared: a connection to a hub. Must operate in half-duplex mode.

a) Edge

- Ports are connected to end hosts.
- Because there's no risk of creating a loop, they can move straight to the forwarding state without the negotiation process.
- They fxn like a classic STP port with Portfast enable.
- If they're in duplex it show Peer-to-Peer edge on cli.

To configure.

SW1 # spanning-tree portfast.

b) Poin-to-Point.

- Connect directly to another switch
- They fxn in full duplex.
- We don't need to configure, switch'll automatically detect if we wanna configure manually.

SW1 (#config-if) # Spanning-tree link-type point-to-point.

c) # Shared.

- Shared Ports connect to ~~each other~~ another switch (or switches) via a hub.
- They fxn in half-duplex.
- You don't need to configure the interface as shared (it should be detected)

~~Sw1 (config-if)#~~
Sw1 (config-if)# spanning-tree link-type shared.

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